

AUTOMATIC HydroLevel REGULATOR

Introduction

- An automatic water motor starter system is designed to manage water pump operation based on water levels.
- It ensures the pump starts and stops automatically, improving efficiency and preventing damage from dry running or overflow.
- This system integrates sensors and control circuits to provide reliable and maintenance-friendly operation for residential and industrial use.

Functionality Overview

- ❑ The system detects water levels through sensors, triggering the motor to start when the water level drops below a preset threshold.
- ❑ It automatically stops the motor when the desired water level is reached, ensuring efficient water usage.
- ❑ This automation prevents pump damage and energy wastage by eliminating manual control.



Level Indication and Alarms

- 5-Level Indicators: Visual representation of the tank level.
- Buzzing: Activates when the tank is filling and reaches level 5.
- Bipping: Activates upon reaching the low level (level 5 is full, so this might mean an alarm for a low-level threshold).
- LED Indicators: An LED blinks when the tank is filling.
- Level Settings: The system allows setting to be set 'at which level motor should ON and OFF automatically.

Operation and Control

- 3-Function Switch: A switch to select between Auto, Manual, or semi automatic modes.
- A switch to BYPASS all automation in extra cases.
- Emergency Switch: A dedicated switch for emergency indication with no other input like mains.
- Motor Control: Auto ON/Auto OFF: Automatic motor operation based on tank levels.
- Automatic Detection of Tank Full: Prevents the motor from turning OFF auto, if motor is ON manually for some purposes
- System Check: An inbuilt 2-minute auto-checking system for water level verification. Purpose to not send electrical signal always.

Power and Safety

- Power Control: Separate Power ON and Power OFF buttons.
- Inbuilt Supply: An internal power supply for system.

OR,

Input Voltage: Compatible with a 5V to 30V input socket. No constant voltage is needed.

- Battery Management: Switch for Charging Battery: A manual switch to initiate battery charging, protecting battery when not needed for long duration.
- Auto Battery Charging: The battery automatically charges when the motor is running.
- Battery Level Indicator: Indicates the battery's charge up to 4 level.

Safety Features

- Fuse for Current Safety: A fuse to protect against overcurrent.
- Auto-Correction Function: when setting is set to be auto ON and OFF level, it can detect an Invalid Input, & Auto Correction Function for auto ON/OFF.
- An additional feature ensure if by any chance or due glitches in circuit motar is not get OFF at desire setting then it will auto off at level 5.

Components and Design

- ❑ Key components include water level sensors, a motor starter relay, control circuitry, and power supply.
- ❑ Sensors monitor water levels, and the control unit processes signals to operate the motor.
- ❑ The design emphasizes durability, accuracy, and safety, suitable for various water tanks and pumping applications.

Installation and Setup

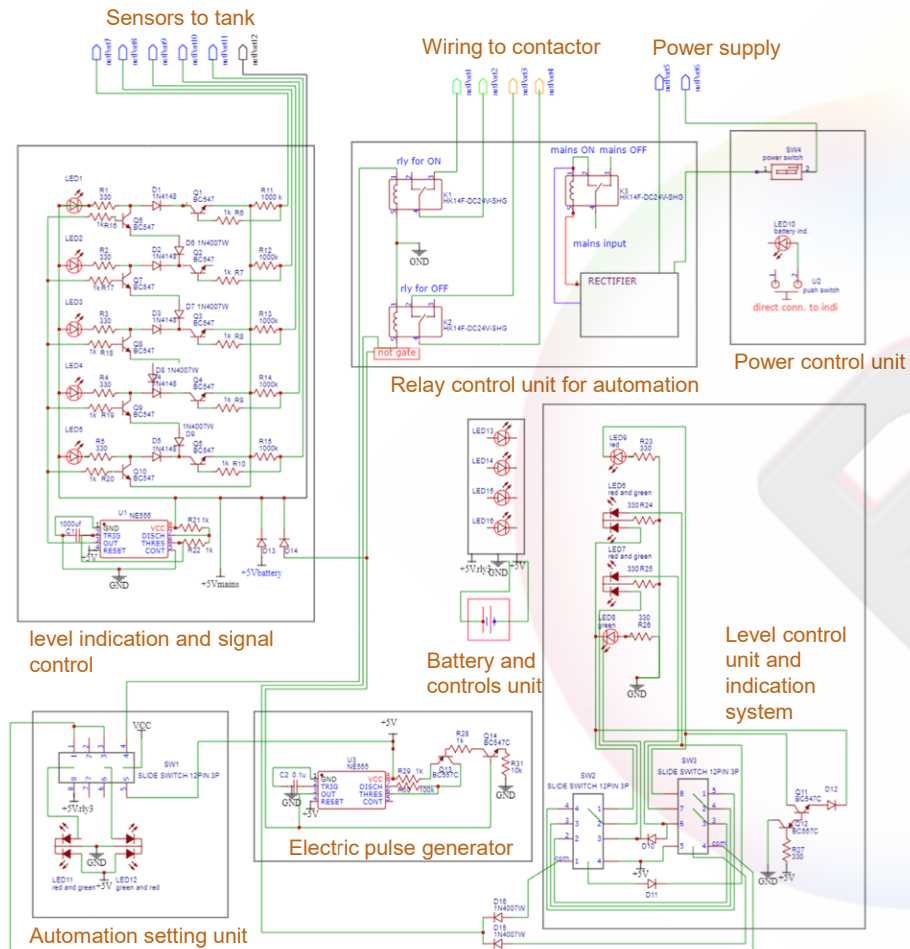
Proper installation is critical for optimal system performance. The motor starter and sensors must be securely mounted in appropriate locations, considering water tank size and pump specifications. Electrical connections should comply with safety standards and include protective devices. Calibration of sensors ensures accurate water level detection. Professional installation minimizes risks and guarantees long-term reliability.



Water Level Auto Control System circuit

Auto On and Auto Off circuit, working & Mechanism

This system automatically starts the water motor when the water level falls below a set point and stops it once the desired level is reached. The mechanism relies on real-time sensor data processed by the control unit, ensuring efficient water management and preventing pump damage due to dry running or overflow.



❑ This circuit diagram illustrates the complete and only important layout of the water level control system.

Disclaimer

For clarity and simplicity, not all internal wiring and interconnections between units are shown. This diagram focuses on functional layout and avoids unnecessary complexity in visual representation.

A. Status Board & control circuit

System control
indication unit

System control
unit

Blinking circuit

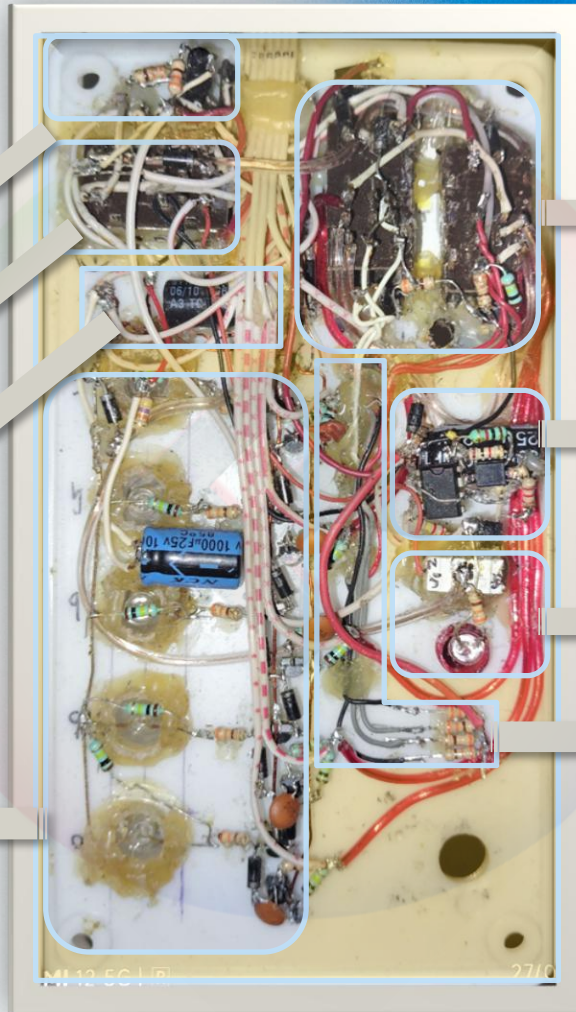
Water level
indicating unit

level control
unit

Signal
generating unit

Battery control
unit

Level control
indication unit



**Real working modal internal circuit
Pic 1 (upper part)**



A. Status Board & control circuit

System control
indication unit

System control
unit

Blinking circuit

Water level
indicating unit



level control
unit

Signal
generating unit

Battery control
unit

Level control
indication unit

*Real working modal internal circuit
Pic 1 (upper part)*



B. Power, automation & Execution Panel

RLY_3

RLY_1

RLY_2

BMS charging module

Switch to control charge



Connector 1

Connector 2

Power switch

Rectifier unit

Buzzer

Battery

Connectors 3

Real working modal internal circuit
Pic 2 (lower part)

WATER LEVEL INDICATING UNIT

- A water level indicating circuit monitors the liquid level in a tank using sensors and displays the status via LED indicators.
- It helps automate control actions like pump ON/OFF based on predefined water levels

BLINKING CIRCUIT

- The blinking circuit activates when the motor is ON and the tank is in the filling state.
- It provides a visual indication that the water level is rising during active operation.

SYSTEM CONTROL UNIT

- The system control unit manages operational modes: Auto, Manual, and Semi Auto.
- Mode selection is achieved through a 3-way slide switch integrated into the unit.
- This setup allows flexible control based on user preference or system requirements.

SYSTEM CONTROL INDICATION UNIT

➤ Motor ON Condition

When the motor is ON, **LED 1 glows green** indicating tank filling has started.

Simultaneously, **LED 2 turns red**, signaling that the **auto-on mechanism has ended** and **auto-off is active**

➤ Motor OFF Condition

When the motor is OFF, **LED 1 glows red**, indicating the **auto-off mechanism has ended**.

At the same time, **LED 2 turns green**, showing that the **auto-on mechanism is now active**

➤ Level Check Pulse

Every **2 minutes**, a signal is sent to the tank to check water level.

During this instant, **led 2 LEDs blink for 1 second** as a visual pulse indicator.

➤ Mode-Based LED Behavior

- In **Semi-Auto Mode**, only **LED 1** is active, showing red or green based on motor state.

- In **Fully Automatic Mode**, both **LED 1 and LED 2** operate as per the above conditions.

➤ Manual Mode (None)

When the switch is set to **Manual (None)**, **no LEDs are active**, and the indication system remains idle.

LEVEL CONTROL UNIT

- This **unit allows users to set custom water levels** for motor ON (auto-on) and motor OFF (auto-off) operations.
- Typical settings include levels 1, 2, 3, 4 for **auto-on** and 5, 4, 3, 2 for **auto-off**, based on tank height.
- The system includes an auto-recognition feature that detects invalid input combinations (e.g., **auto-on < auto-off**).
- In such cases, it automatically corrects the input—for example, if auto-off is set to 3 and auto-on to 3 or 4, it resets **auto-on to level 2**.
- An **LED** indication confirms the correction, ensuring safe and logical operation without manual intervention.

SIGNAL GENERATING UNIT

- The signal generating unit **emits a pulse every 2 minutes** to assess the water level in the tank.
- This **signal triggers the sensing mechanism**, enabling the control system to respond accordingly.
- It ensures timely updates for accurate level monitoring and automated decision-making.

BATTERY CONTROL UNIT

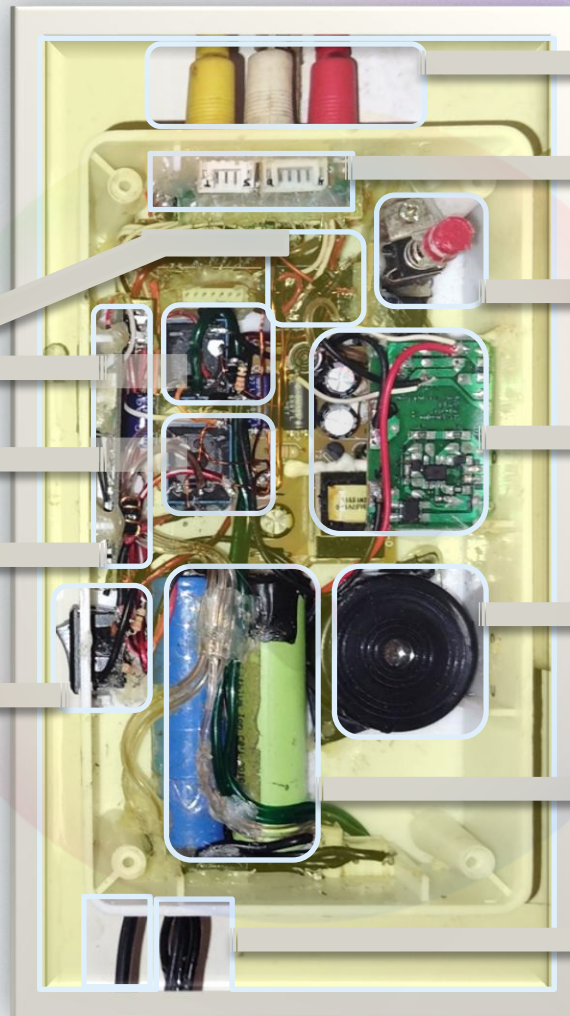
- This unit enables water level indication during power outages or poor weather conditions.
- It includes a **4000mAh** rechargeable battery, a charging module, and a push switch for manual activation.
- Users can check tank status without mains supply by pressing the switch to trigger LED indicators.
- **Ideal for emergency scenarios**, ensuring uninterrupted monitoring when grid power is unavailable.

LEVEL CONTROL INDICATION UNIT

- This unit **visually displays** using **LED** indicator the user-defined **ON** and **OFF** levels using colored LEDs.
- **Green LED indicates** the **auto-on** level and is positioned lower than the red LED, which marks the **auto-off** level.
- It also reflects the auto-correction logic—preventing invalid settings where **ON level < OFF level**.
- If such an error is detected, the system auto-adjusts the ON level and updates the LED indication accordingly.

B. Power, automation & Execution Panel

RLY__ 3
RLY__ 1
RLY__ 2
BMS  charging module
Switch to control charge



Connector 1
Connector 2
Power switch
Rectifier unit
Buzzer
Battery
Connectors 3

*Real working modal internal circuit
Pic 2 (lower part)*

RLY__ 1

- ❑ Relay__ 1 is responsible for triggering the automatic shut-off function. It activates to auto-off system when the predefined condition that is set by “level control unit” is met.

RLY__ 2

- ❑ Relay 2 triggers and automatic power-on the mechanism when the defined condition by “level control unit” is met.

RLY__ 3

- ❑ Relay 3 is responsible for sending control signals to the starter box. It enables ON and OFF operations based on commands from the control unit. This relay ensures reliable switching and system responsiveness

CONNECTOR 1

- ❑ It is used to establish a reliable link between the control unit and tank sensors. It ensures accurate signal transmission for responsive system behavior.

CONNECTOR 2

- ❑ Connector 2 is responsible for establishing the internal connection within the control unit. It links the upper and lower sections, ensuring unified signal flow and operational integrity.

CONNECTOR 3

- ❑ Connector 3 is responsible for transmitting control signals to the starter box. It enables the activation and deactivation of the system via ON/OFF commands. This ensures reliable communication between the control unit and the starter mechanism.

Conclusions

An automatic water motor starter with water level control enhances operational efficiency by automating pump operation. Careful installation, accurate sensor calibration, and regular maintenance are essential for reliable performance. This system safeguards the motor, conserves water, and reduces manual intervention, making it a valuable solution for water management in various settings.

